

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B.Tech

Branch : Electrical and Power/Electrical & Electronics

Engineering/Electrical Engineering

Subject Code & Name: BTEEC503 & Power Electronics

Semester : V

Max Marks: 60

Date:11/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

**Q. 1 Objective type questions. (Compulsory Question)**

**12**

- |   |                                                                                                                                   |                                                                                                        |                                                                                                       |                                                                                                              |
|---|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| 1 | The terminal characteristics of power electronics devices like SCR and MOSFET refer to:                                           | <b>CO1</b>                                                                                             | 1                                                                                                     |                                                                                                              |
|   | a. The power consumption of the device                                                                                            | b. The voltage and current relationships at the device terminals                                       | c. The thermal properties of the device                                                               | d. The mounting specifications                                                                               |
| 2 | Which of the following is primarily used for protection in power electronics circuits?                                            | <b>CO1</b>                                                                                             | 1                                                                                                     |                                                                                                              |
|   | a. Firing circuits                                                                                                                | b. Snubber circuits                                                                                    | c. Cooling systems                                                                                    | d. Mounting systems                                                                                          |
| 3 | Which of the following statements is most accurate regarding the gate turn-off (GTO) thyristor in comparison to the standard SCR? | <b>CO1</b>                                                                                             | 1                                                                                                     |                                                                                                              |
|   | a. GTO can be turned off by applying a negative gate current, whereas SCR requires the current to drop to zero for                | b. GTO has a slower switching speed than SCR, making it less suitable for high-frequency applications. | c. SCR has superior thermal performance, whereas GTO suffers from significant thermal runaway issues. | d. GTO is typically used in applications with very low voltage requirements due to its high turn-on voltage. |

|    |                                                                                                                                                                                                                              |     |   |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---|
|    | turning off.                                                                                                                                                                                                                 |     |   |
| 4  | The DC output voltage in phase-controlled rectifiers depends on:<br>a. Only the frequency of the AC supply<br>b. The firing angle<br>c. The inductance of the load<br>d. The size of the transformer                         | CO2 | 1 |
| 5  | Which of the following is a major application of choppers?<br>a. AC motor drives<br>b. DC motor drives<br>c. AC voltage regulation<br>d. Frequency synthesis                                                                 | CO3 | 1 |
| 6  | The harmonic reduction in the inverter output voltage can be achieved by:<br>a. Using filters<br>b. Increasing the modulation frequency<br>c. Using PWM techniques<br>d. All of the above                                    | CO3 | 1 |
| 7  | AC voltage controllers are primarily used to:<br>a. Control the DC output voltage<br>b. Regulate the AC input voltage to an appropriate level<br>c. Convert AC to DC<br>d. Improve the efficiency of inverters               | CO4 | 1 |
| 8  | In a DC-DC buck converter, if the input voltage is 24V and the duty cycle is 40%, what will be the output voltage?<br>a. 9.6V<br>b. 12V<br>c. 10.4V<br>d. 7.6V                                                               | CO3 | 1 |
| 9  | In a single-phase controlled rectifier with an RL load, the AC supply voltage is 230V RMS. If the firing angle is $90^\circ$ , calculate the average DC voltage across the load.<br>a. 110V<br>b. 120V<br>c. 150V<br>d. 100V | CO2 | 1 |
| 10 | What is the main function of a cycloconverter?<br>a. Convert AC to DC<br>b. Convert DC to AC<br>c. Convert AC to AC with variable frequency<br>d. Convert AC to DC with variable frequency                                   | CO4 | 1 |
| 11 | Which harmonic component is primarily targeted by a well-designed filter in an inverter system?<br>a. DC component<br>b. Fundamental frequency<br>c. Odd harmonics<br>d. Even harmonics                                      | CO3 | 1 |
| 12 | The inverters are primarily used for                                                                                                                                                                                         | CO3 | 1 |

- |                           |                           |                           |                           |
|---------------------------|---------------------------|---------------------------|---------------------------|
| a. AC to AC<br>conversion | b. DC to DC<br>conversion | c. DC to AC<br>conversion | d. AC to DC<br>conversion |
|---------------------------|---------------------------|---------------------------|---------------------------|

**Q. 2 Solve the following.**

12

- A) Describe the terminal characteristics of following power electronic devices.

CO1

6

- a) SCR  
b) BJT  
c) MOSFET

- B) What is TRIAC? Explain turning on process and V-I characteristics of TRIAC.

CO1

6

**Q.3 Solve the following.**

12

- A) Elucidate with a neat circuit diagram working of single phase half wave controlled rectifier for R load, Draw necessary waveforms. Derive an expression for output voltage ( $V_o$ ), output current ( $I_o$ ).

CO2

6

- B) A single phase-controlled rectifier bridge consists of one SCR and three diodes as shown in the figure below.

CO2

6

- a) Sketch output voltage waveform for a firing angle  $\alpha$  for the SCR and hence obtain an expression for the average output voltage under the assumption of continuous current. Show the conduction of various components as well.

- b) For an AC source voltage 230V, 50Hz and firing angle of  $45^\circ$ , find average output voltage and, find average output current in case load consist of  $R=5\Omega$ ,  $L=8\text{mH}$  and  $E=100\text{V}$ .

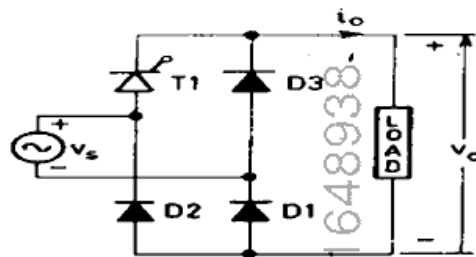


Fig. circuit diagram of controlled rectifier

**Q. 4 Solve Any Two of the following.**

12

- A) Discuss the different control strategies used in chopper circuits for regulating the output voltage. Compare time ratio control and current

CO3

6

limit control strategy of chopper.

- |                                                                                                                                                                                                                                                           |     |   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---|
| B) Illustrate with a neat circuit diagram working of a four quadrant or type E chopper.                                                                                                                                                                   | CO3 | 6 |
| C) For Type A chopper as shown in figure, DC source voltage=230V, load resistance =10Ω. Take voltage drop of 2V across chopper when it is on. For a duty cycle of 0.4 calculate:<br>a) average and rms values of output voltage<br>b) chopper efficiency. | CO3 | 6 |

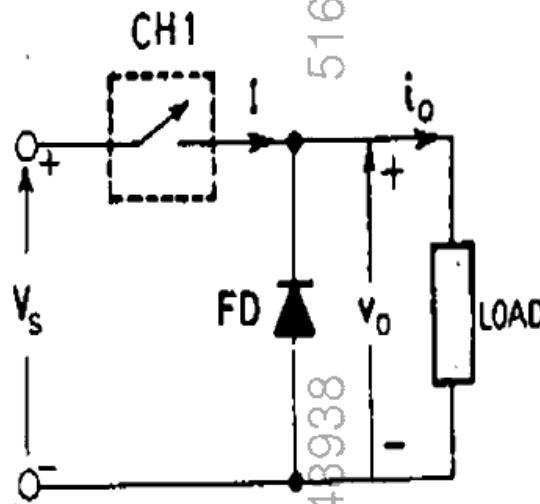


Fig. Circuit diagram of Type A Chopper

- |                                                                                                                                             |     |    |
|---------------------------------------------------------------------------------------------------------------------------------------------|-----|----|
| <b>Q.5 Solve Any Two of the following.</b>                                                                                                  |     | 12 |
| A) With a neat circuit diagram explain working of a single-phase full bridge inverter for R load. Draw necessary waveforms.                 | CO3 | 6  |
| B) Define the term harmonics? What are the methods for harmonics reduction in inverter output voltage. Illustrate any one method in detail. | CO4 | 6  |
| C) What is PWM technique? Mention different types of PWM technique. Elucidate in detail SPWM technique.                                     | CO4 | 6  |

- |                                                                                                                                        |     |    |
|----------------------------------------------------------------------------------------------------------------------------------------|-----|----|
| <b>Q. 6 Solve Any Two of the following.</b>                                                                                            |     | 12 |
| A) Elucidate working of single-phase full-wave AC voltage controller with the help of power circuit diagram. Draw necessary waveforms. | CO4 | 6  |

- B) A single-phase half-wave AC voltage controller feeds a load of  $R=20\Omega$  with an input voltage of 230V, 50Hz. Firing angle of thyristor is  $45^\circ$ . Determine,
- a) rms value of output voltage
  - b) Power delivered to load
- C) What is cycloconverter? Explain step-down cycloconverter with a neat circuit diagram. Draw necessary waveforms.

CO4

6

CO4

6

\*\*\* End \*\*\*